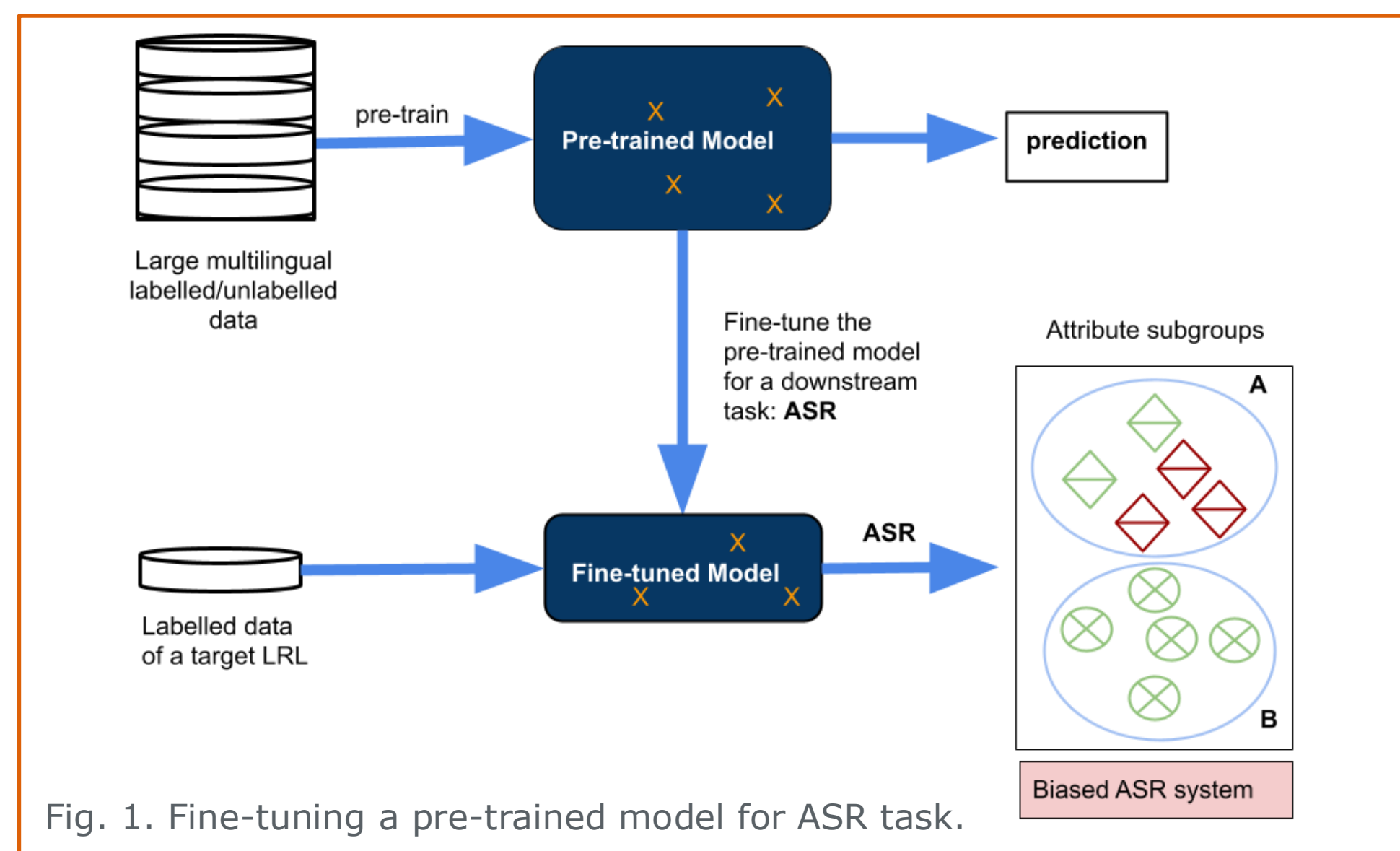


Mitigating Speaker Attribute Bias in Automatic Speech Recognition Models for Low Resource African Languages

Introduction

Attribute-specific bias remains a challenge in automatic speech recognition (ASR) systems. If not mitigated, bias has the potential to limit certain groups of people from accessing speech technology systems based on demographic information such as gender, age and speaker type.



Aim

To develop a fine-tuning algorithm that minimizes speaker attribute bias in speech recognition systems for low resource languages, while preserving speech recognition accuracy.

Research Questions

- To what extent do speech recognition models fine-tuned from multilingual pre-trained models exhibit attribute-specific bias in the target languages?
- To what extent does debiasing pre-trained speech models using adapters via adapter fusion minimize attribute-specific bias in ASR models in the target languages?
- To what extent does coupling group distributionally robust optimization with domain adversarial training (DRODAT) minimize speaker-attribute bias in fine-tuned ASR models in the target languages?

Methodology

- Attribute-specific bias:** gender & speaker type (native vs. Non-native)
- Target languages:** African languages: Bemba & Nyanja
- Proposed method:** Group distributionally robust optimization with domain adversarial training (DRODAT)
- Pretrained speech models:** MMS, Whisper, SSA-HuBERT-X-60K

Data Collection: Bias Evaluation Dataset

Objective: To create a bias evaluation dataset to examine speaker-attribute bias in speech recognition models for our target languages.

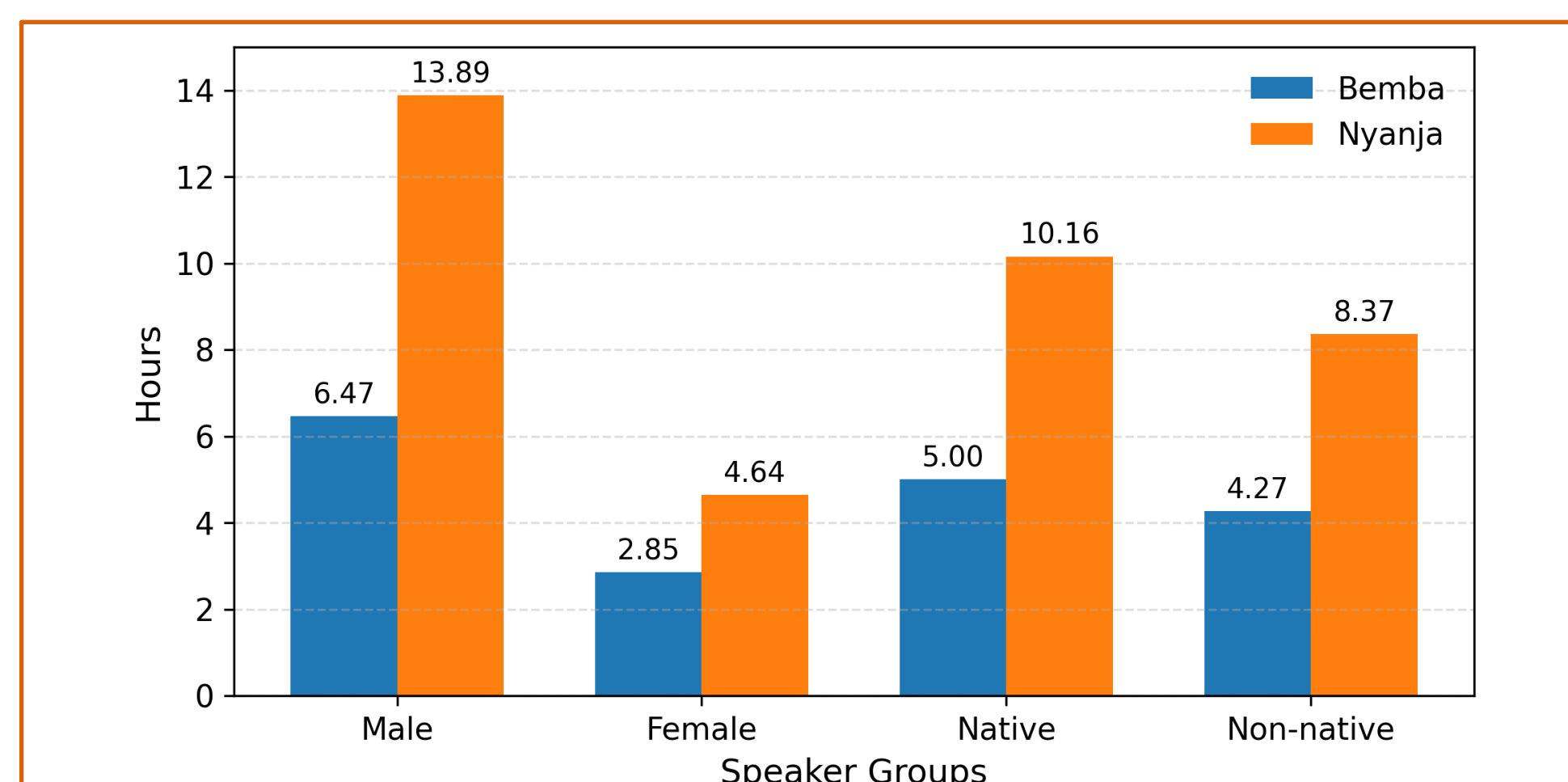


Fig. 2. Results of the data collection project conducted to create a bias evaluation for our target languages.

Phase 1: Bias Analysis

Objective: To investigate the impact of fine-tuning multilingual pre-trained speech models on gender bias in ASR systems for three low resource African languages: **Bemba, Nyanja** and **Swahili**.

Results: There is bias transfer from pre-trained models during fine-tuning [1].

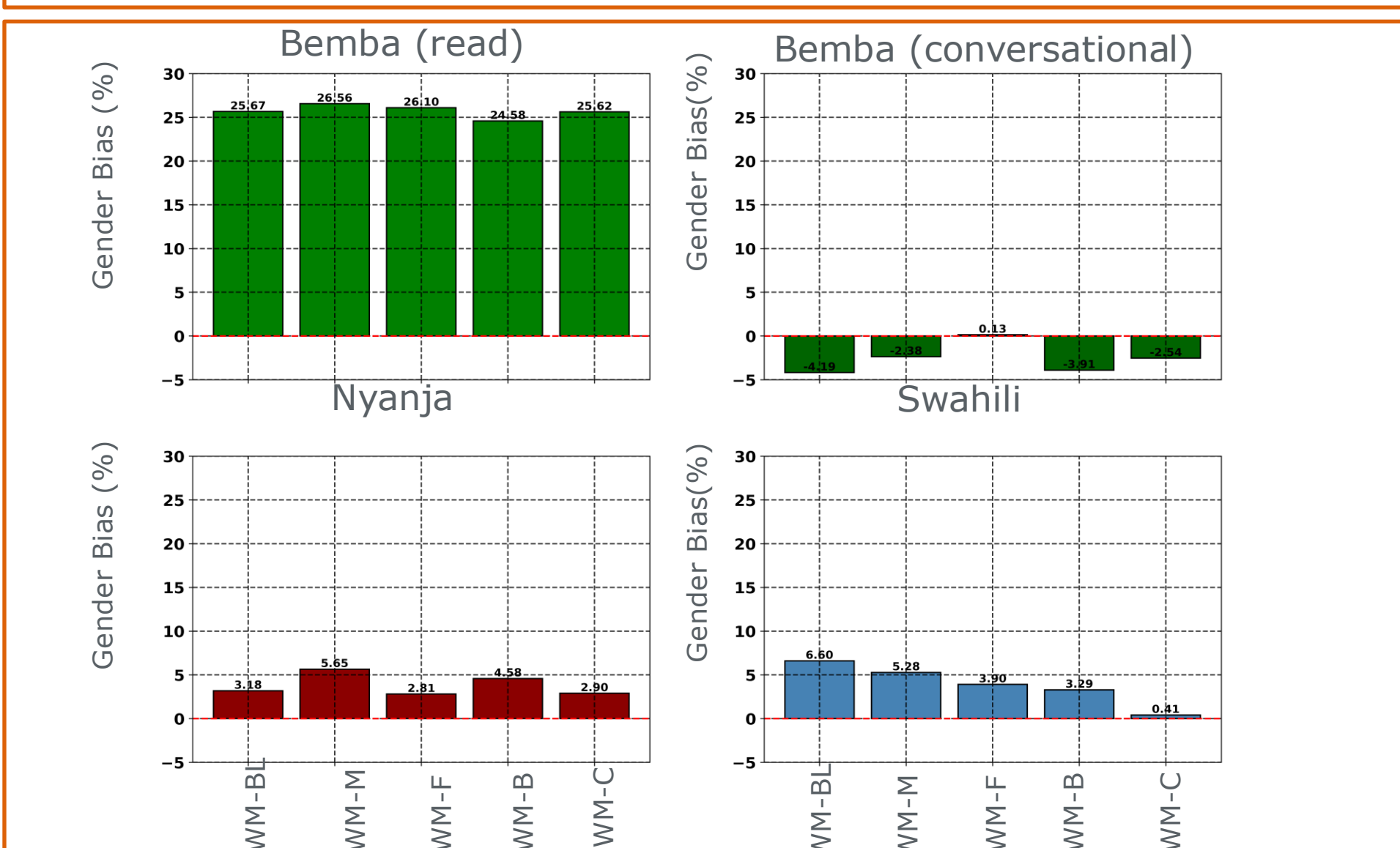


Fig. 3. Gender bias (%) estimates in fine-tuned Whisper models (WM).

Phase 2: AdapterFusion

Objective: To investigate the extent to which fusion of ASR-specific (trained with CTC) and Debiasing (trained with DAT) adapters using an AdapterFusion module minimizes speaker-attribute bias in fine-tuned speech recognition models for Bemba and Nyanja.

Results: Based on experimental results, we found that using the AdapterFusion-based method does not reduce speaker-attribute bias (gender) in fine-tuned speech recognition models for Bemba and Nyanja.

Phase 3: DRODAT

Objective: To investigate the extent to which the fine-tuning method that couples group distributionally robust optimization and domain adversarial training (DRODAT) minimizes speaker-attribute bias in fine-tuned speech recognition models for Bemba and Nyanja.

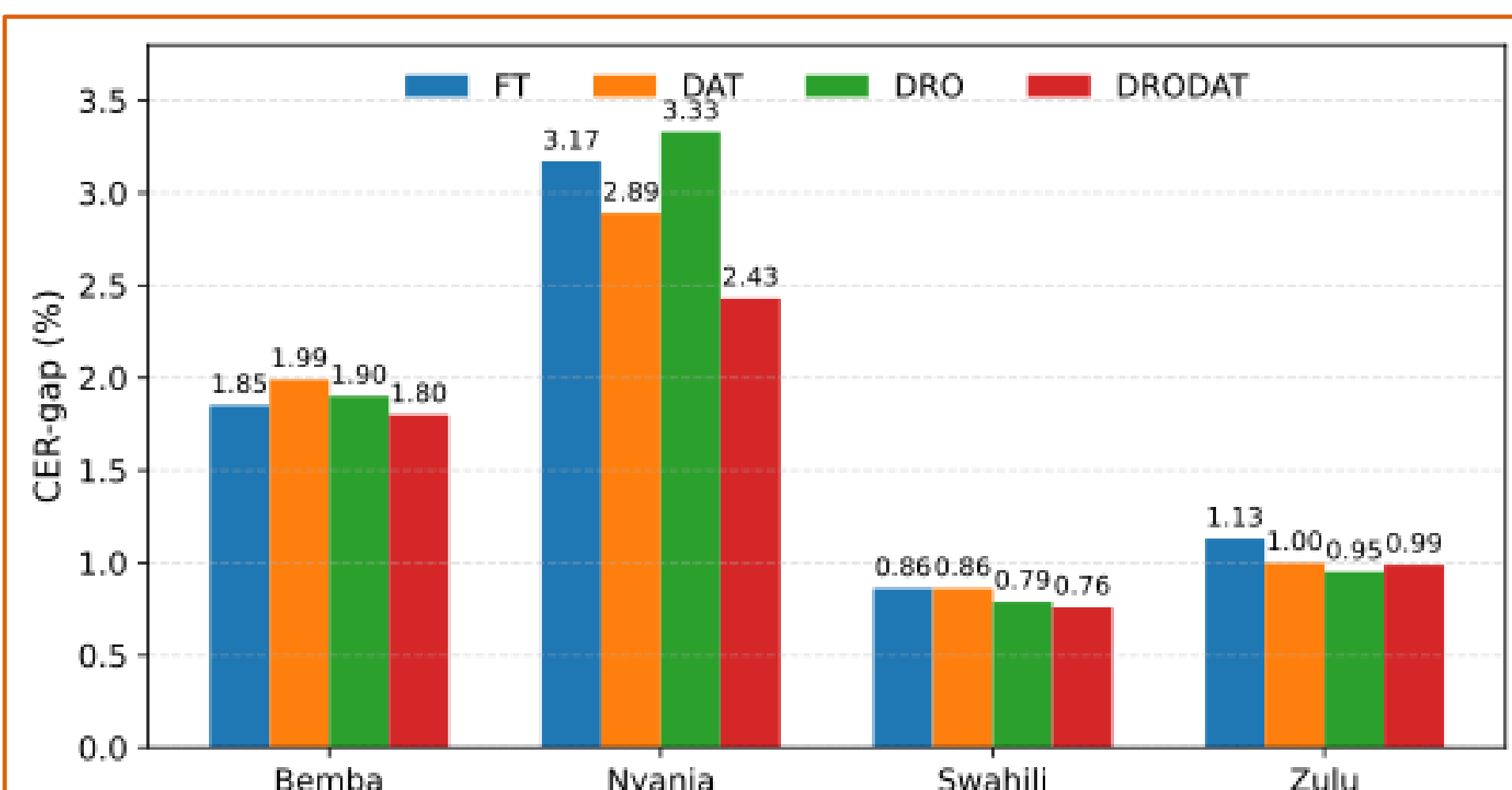


Fig. 4. Our proposed method, DRODAT-based achieves greater bias reduction in Bemba, Nyanja, and Swahili.

References

- [1] C. Sikasote, H. Suleman, and J. Buys, "Investigating the impact of multilingual pre-trained speech models on gender bias in asr for low-resource african languages, in Artificial Intelligence Research (A. Gerber and A. W. Pillay, eds.), (Cham), pp. 144–158, Springer Nature Switzerland, 2026

Claytone Sikasote

- Email:** SKSCLA001@myuct.ac.za
- Website:** <https://csikasote.github.io>
- Supervisors:** Dr. Jan Buys and Prof. Hussein Suleman